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number of moles of ethane $= \frac{2 \text{ mol ethane}}{4 \text{ mol } CO_2} \times 1 \text{ mol } CO_2$

$= \frac{1}{2}$ mole of ethane

= 0.5 mole of ethane

41.

$$\overset{-1}{H_2O_2} + Fe^{2+} \longrightarrow Fe^{3+} + \overset{-2}{H_2O}$$

$(1e^- \times 2)$ $1e^-$

$1\ H_2O_2 + 2Fe^{2+} \rightarrow Fe^{3+} + H_2O$

$\Rightarrow \quad H_2O_2 + 2Fe^{2+} + 2H^+ \rightarrow 2Fe^{3+} + 2H_2O$

42.

Element	Percentage	Atomic mass	Relative number of atoms	Simple ratio	Whole no.
C	76.6	12	$\frac{76.6}{12} = 6.38$	$\frac{6.38}{1.06} = 6$	6
H	6.38	1	$\frac{6.38}{1} = 6.38$	$\frac{6.38}{1.06} = 6$	6
O	17.02	16	$\frac{17.02}{16} = 1.06$	$\frac{1.06}{1.06} = 1$	1

Empirical formula = C_6H_6O

$n = \frac{\text{Molar mass}}{\text{Calculated empirical formula mass}}$

$= \frac{2 \times \text{vapour density}}{94} = \frac{2 \times 47}{94} = 1$

$\therefore$ molecular formula $(C_6H_6O) \times 1 = C_6H_6O$